

personalized
by you

SoloSpeak

Personalized Wake-Word Detection for Samsung Devices

custom phrase

+

enrolled voice

→

safer wake-up

on-device
privacy



Team SoloSung

Pranav Angrish

B.Tech Computer Science & Engineering · Year 2026

Thapar Institute of Engineering and Technology

Roll No. 102216085 · pangrish_be22@thapar.edu

voice
identity



Built, trained, evaluated, exported · Stage 7 deployable



Designing a Robust AI System for Speech Disentanglement

Today's devices listen for the phrase —
not the person behind it.

A trigger phrase can be copied.
A similar phrase can confuse.
A voice identity cannot be ignored.

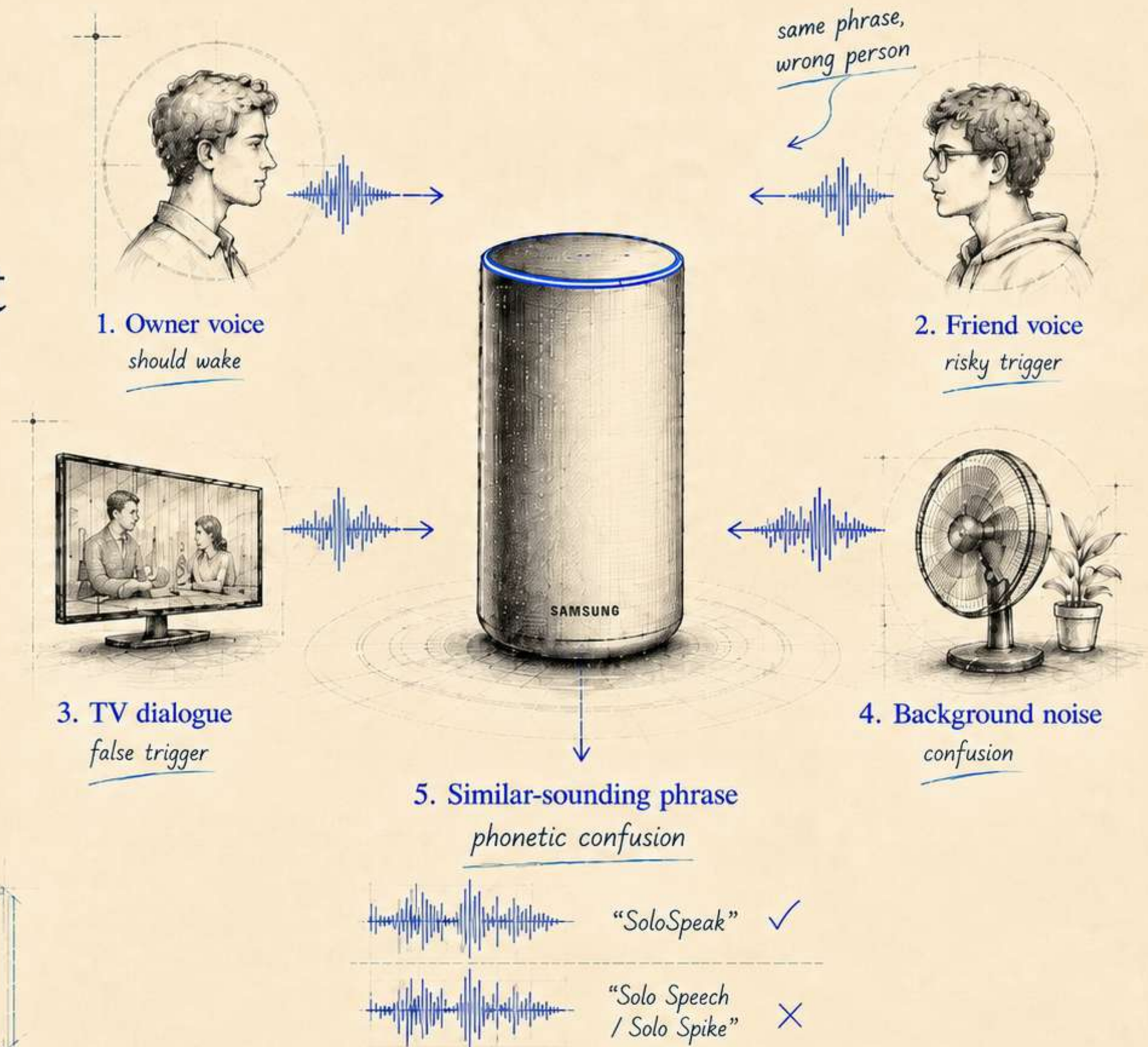
⊗ Today: "Was the phrase spoken?"



👤 Needed: "Was it spoken by the right person?"



Problem insight: Phrase detection alone creates false wake-ups from wrong speakers, similar-sounding phrases, and ambient audio.



Enroll Once. Wake Only for You.

The user speaks their custom phrase 4–5 times. SoloSpeak converts those samples into personal voice templates.

Instead of storing raw voice audio, SoloSpeak stores compact embeddings that represent the enrolled user's phrase and voice identity.

① 1. Enroll User repeats the wake phrase 4–5 times



② 2. Store Templates Embeddings are saved on-device



③ 3. Verify Incoming audio is matched against enrolled templates

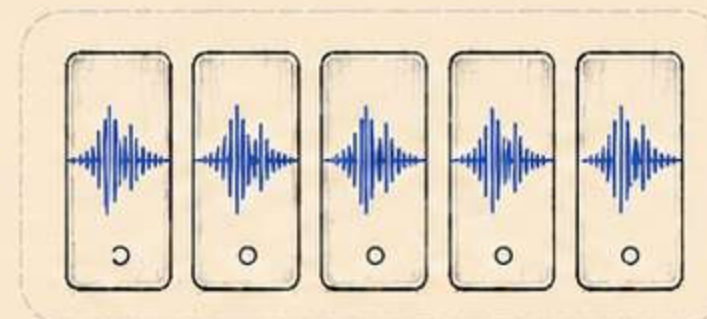


Solution insight: A copied phrase is not enough — the voice must match the enrolled templates.

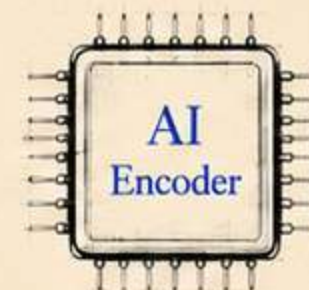
① 1. Owner speaks
custom wake phrase



② 2. Enrollment samples
4–5 captures



③ 3. Encode
AI encoder converts
samples to embeddings



④ 4. Enrolled templates
Stored securely on-device



⑤ 5. Future match
Compare incoming
audio to templates



✓ Match → Verified wake-up
Authorized voice

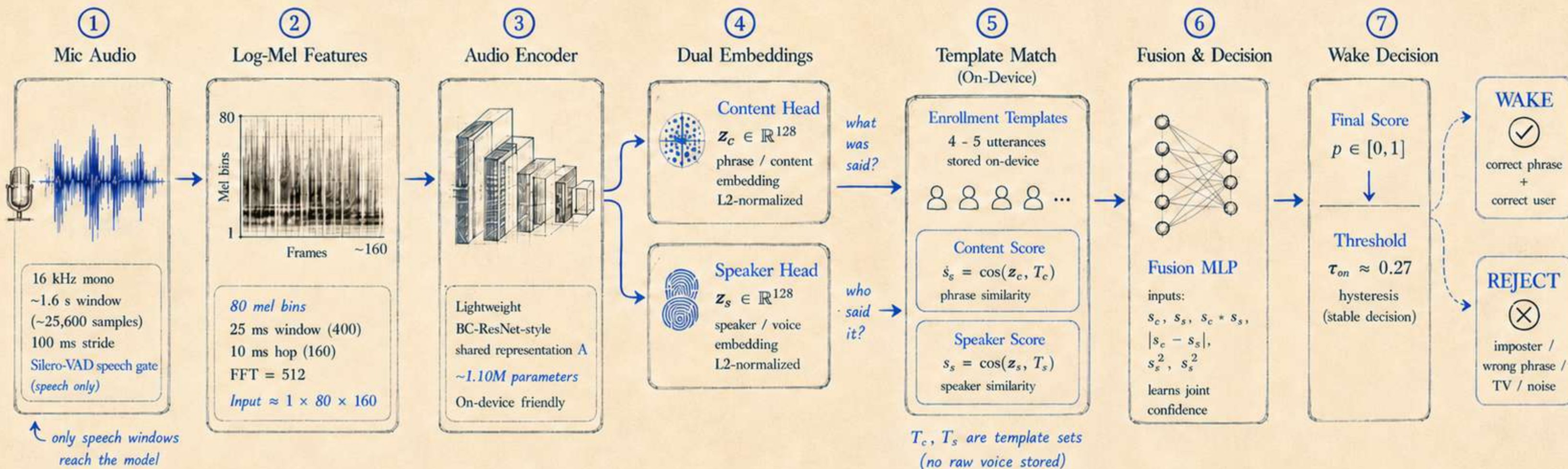
✗ No match → Reject
Unauthorized voice



Runtime Inference Architecture

From live audio to phrase + speaker verified wake decision

live audio
in real time



Key idea: SoloSpeak does not directly classify a wake word. It verifies *both the phrase identity and the speaker identity* and wakes only when both match.



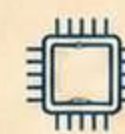
Dual Verification

Checks what was said and who said it.



Privacy First

Templates only. No raw voice uploaded.



Built for Devices

Lightweight model. Fast, local, and secure.

One model. Continuously improved.

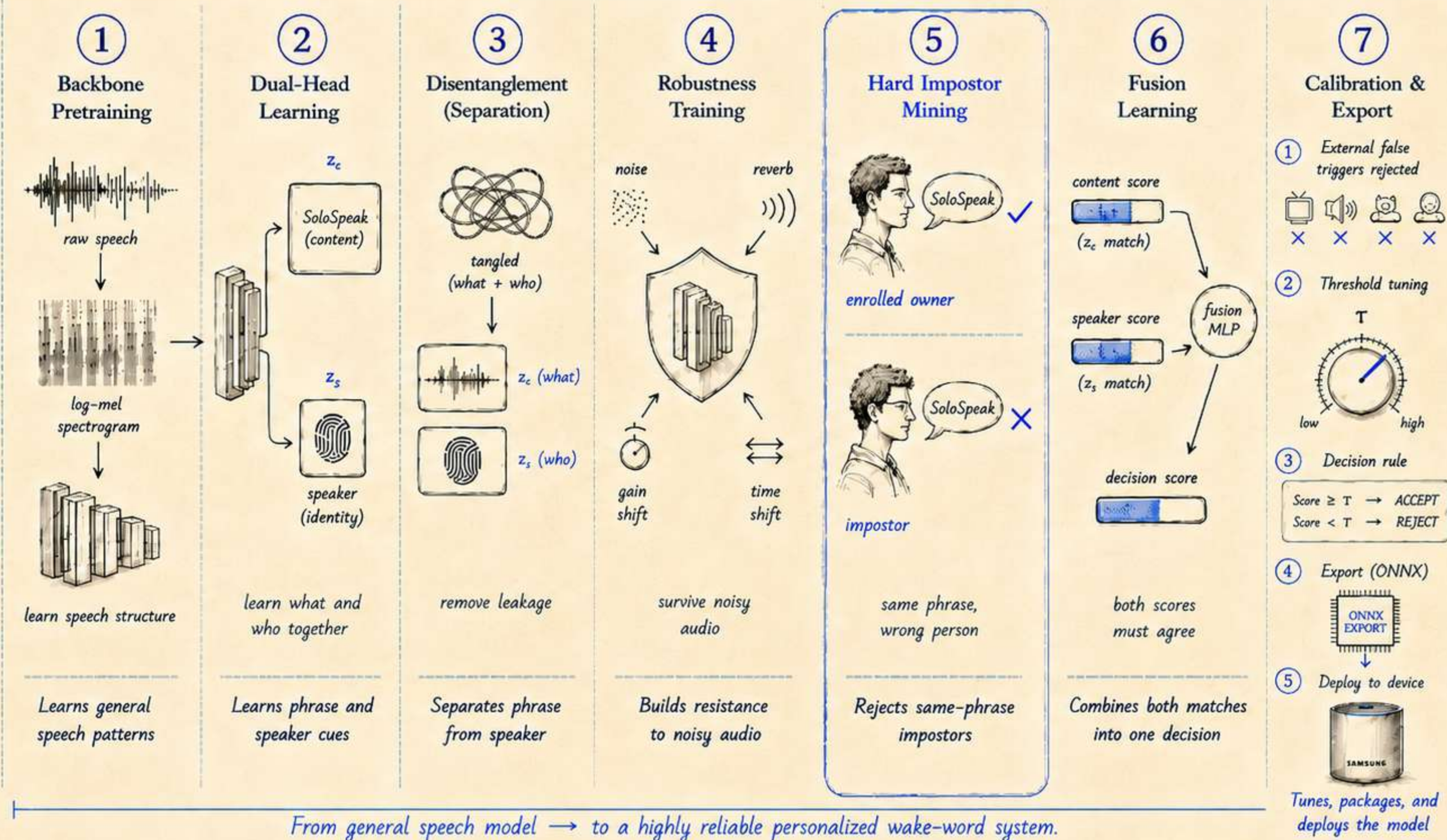
Training SoloSpeak Through 7 Reliability Layers

We didn't train once. We trained against real failure modes — layer by layer — until the model could separate phrase from speaker, resist noisy audio, reject impostors, and deploy reliably on-device.



Key insight:

Each stage adds one capability and removes a specific reason for false wake-ups.



Output: Stage 7 deployable personalized wake-word model

z_c = content embedding
 z_s = speaker embedding
 eg. = wrong speaker + correct phrase
 T = decision threshold

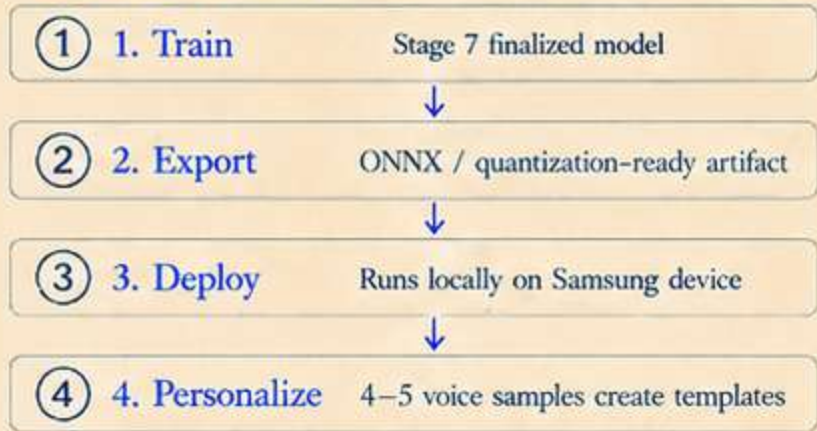
Deploy Once. Personalize Everywhere.

From Stage 7 export to on-device inference across the Samsung ecosystem

A single Stage 7 SoloSpeak model scales across Samsung devices, while each user adds only private on-device templates.

From trained model to real device

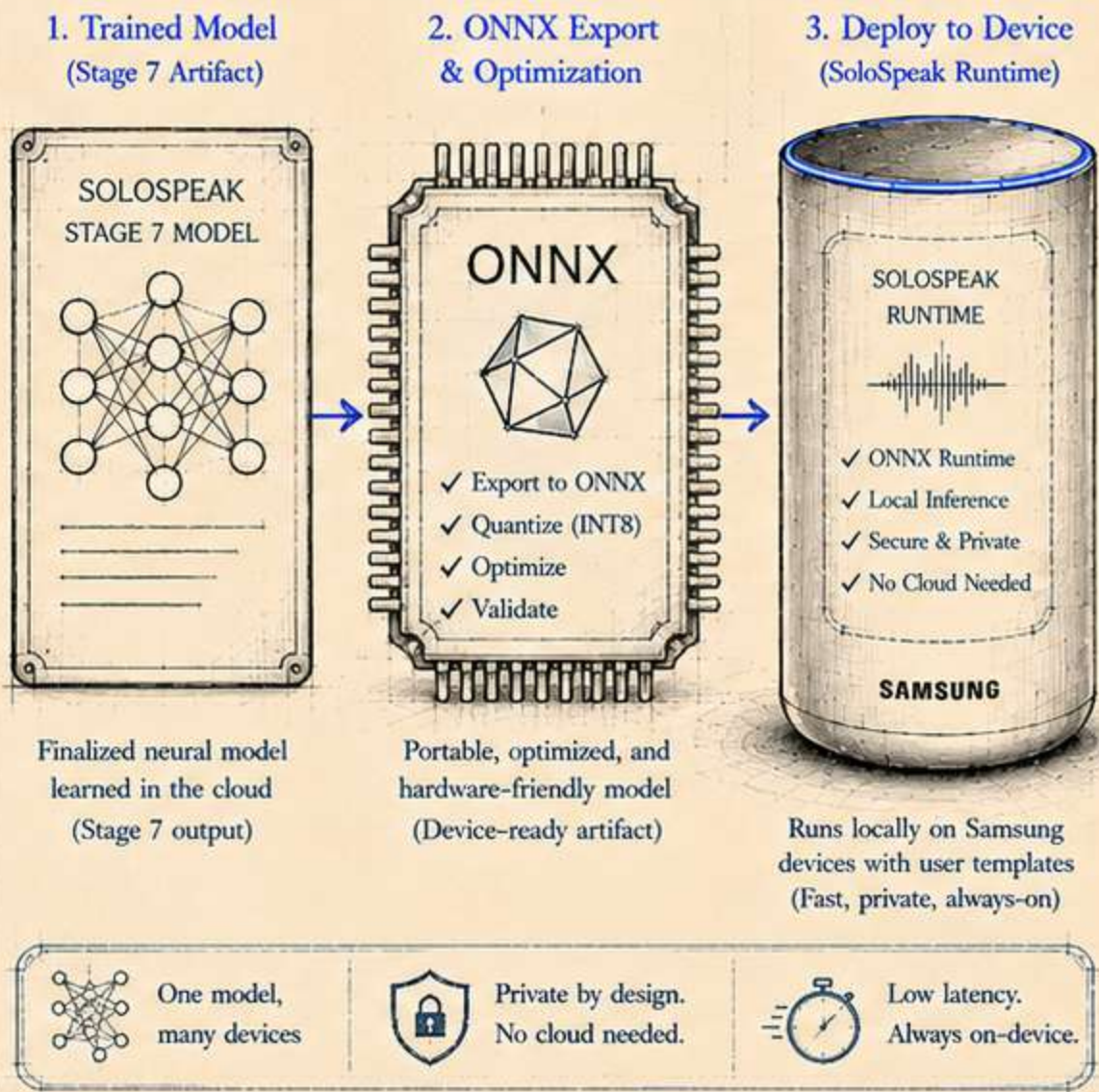
SoloSpeak is trained once, exported as a lightweight deployable model, and placed on-device. Personalization happens later through enrollment templates — not by retraining the model for every user.



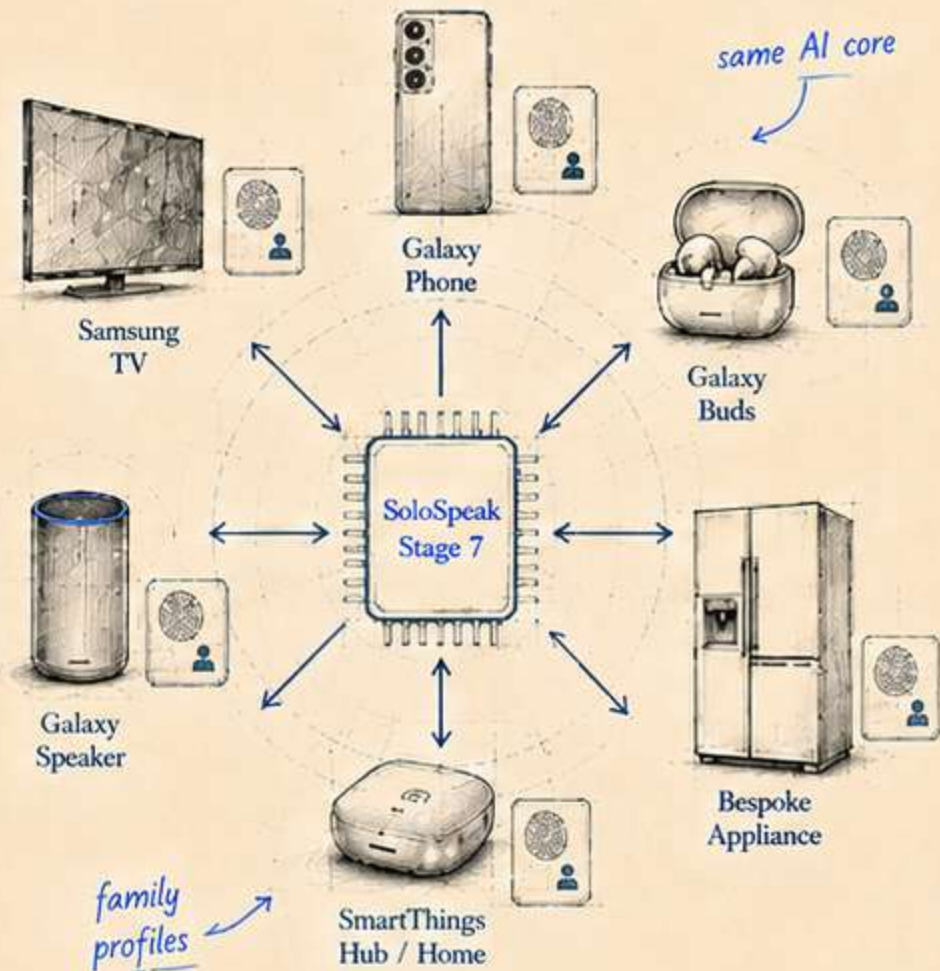
*Deployment insight: The model is shared.
The identity is personal.
The decision stays on-device.*

SoloSpeak Runtime

From trained model to device-ready deployment



Scales across Samsung ecosystem



- ✓ Same base model
- ✓ Different user templates
- ✓ New users enroll — model does not retrain

Private
Templates stay on-device.
No raw voice upload.

Efficient
Speech gate skips silence and non-speech.

Scalable
One base model supports many devices and many users.

Updatable
OTA model updates improve the base system without rebuilding profiles.

Open Data Usage Map

Public datasets train the base model.
Personal profiles come only from user enrollment.

Open data trains the base intelligence. User profiles are built later from the owner's 4–5 enrollment recordings.

Model Learning

① Google Speech Commands v2



Role: keyword / phrase learning
Used in: Stage 1 pretrain + Stage 2 content head
Not used for: user profiles

② VoxCeleb1



Role: speaker identity variation
Used in: speaker head + disentanglement + hard negatives
Helps reject: wrong person saying the phrase

③ LibriSpeech



Role: multi-speaker speech diversity
Used in: speaker learning + external speech trials
Helps build: stronger speaker embeddings

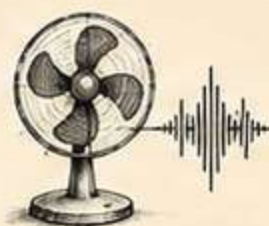
Reliability & Calibration

④ Common Voice



Role: unseen speech false-accept testing
Used in: Stage 7 calibration
Helps tune: final threshold

⑤ UrbanSound8K + BG Noise



Role: environmental trigger stress
Used in: Stage 7 external FA testing
Helps reject: noise, ambience, appliances

⑥ MUSAN + OpenSLR RIRs



Role: noise + room reverb augmentation
Used in: robustness training extension
Helps simulate: real Samsung home environments



User Enrollment Samples

4–5 recordings by the device owner
Used in: enrollment, not base training
Creates: compact on-device templates

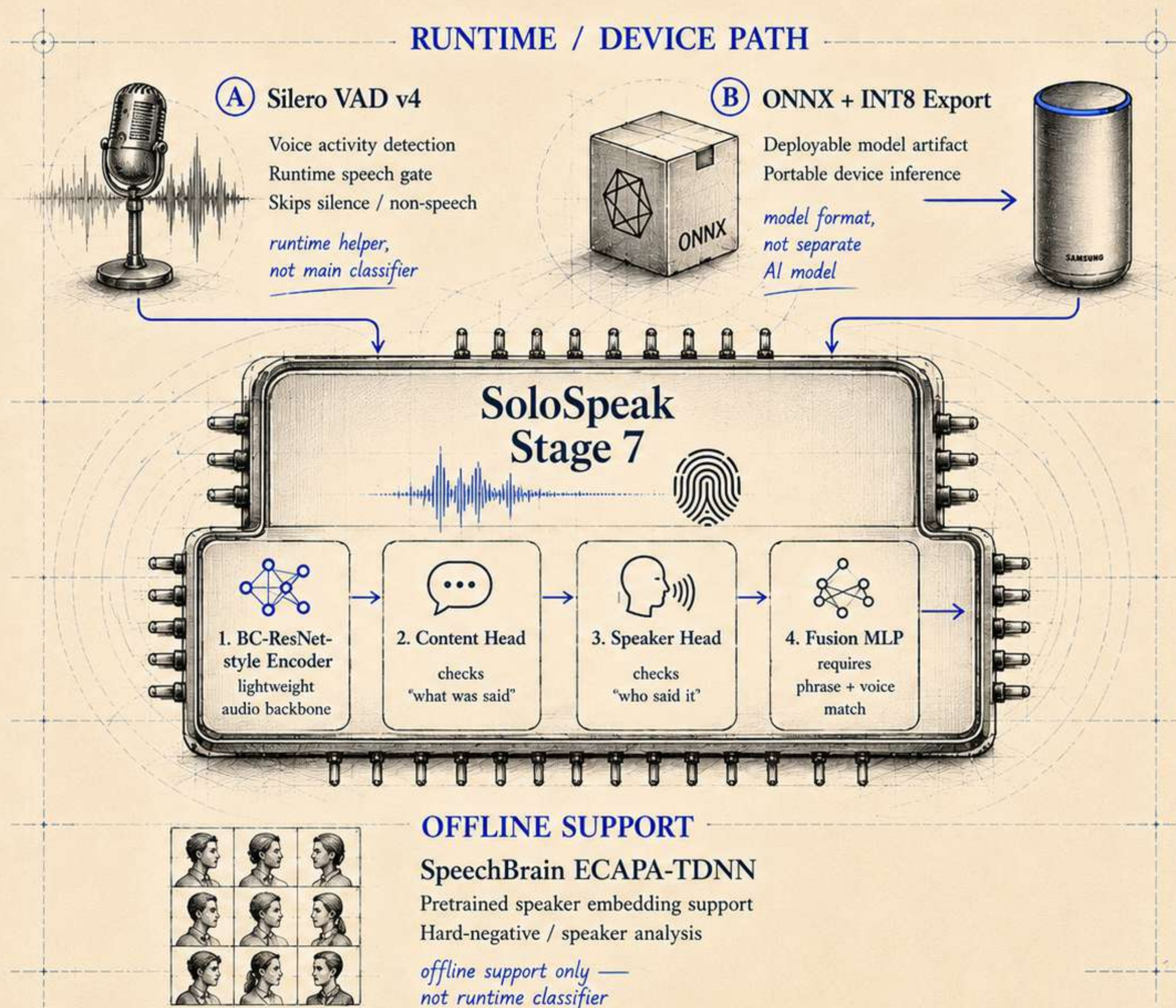


Open data trains the general model. Enrollment audio creates the personal profile. Runtime matching combines both.

Open Model Usage Map

Open-source models support the pipeline.
SoloSpeak makes the final wake decision.

*Open models assist filtering,
speaker analysis, and deployment.
SoloSpeak Stage 7 performs the
final personalized wake decision.*

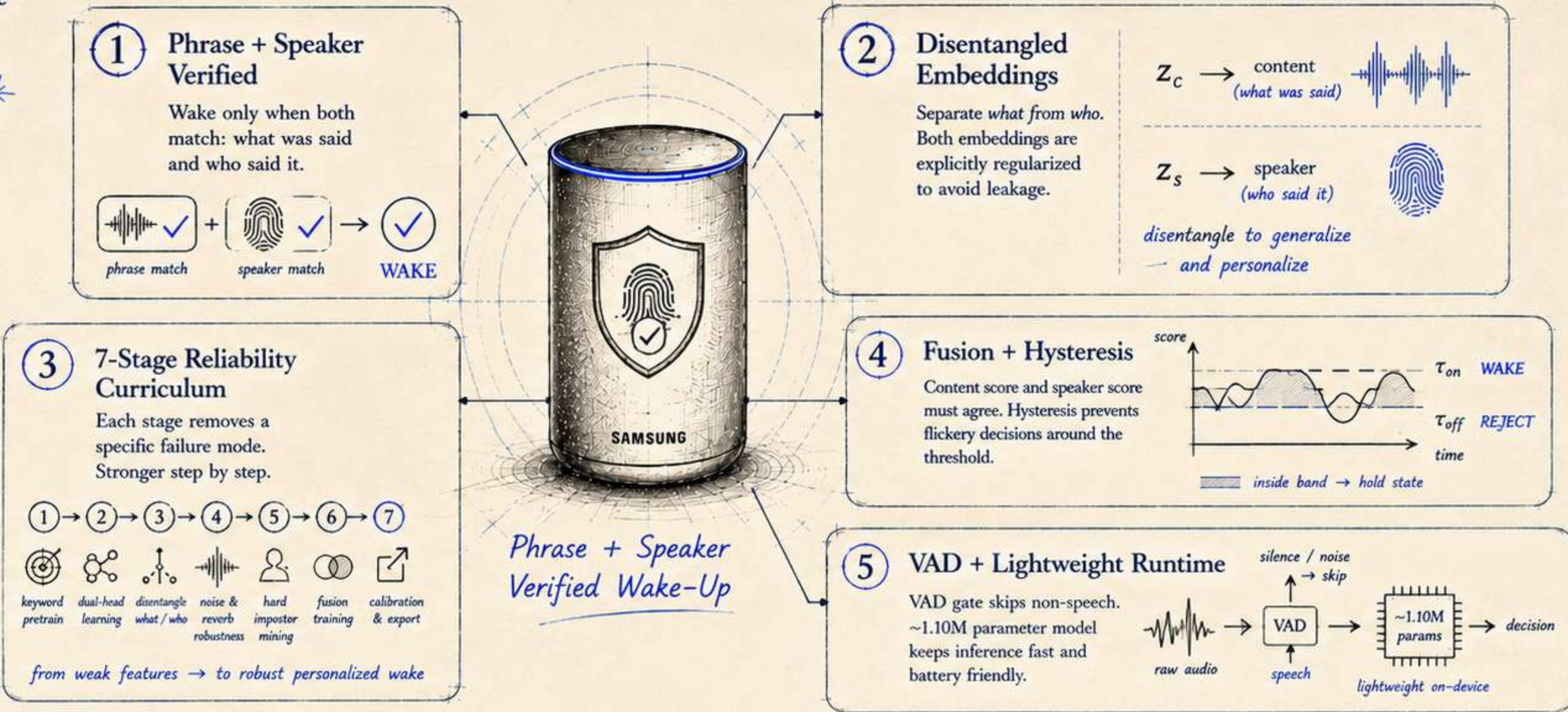


Why SoloSpeak Wins

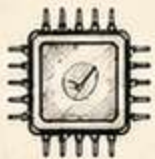
*personalized wake-up,
engineered end-to-end*

SOTA gives strong keyword detectors and speaker verifiers.
SoloSpeak turns them into a deployable personalized wake system:
disentangled phrase/speaker embeddings, hard-impostor training, stable hysteresis, VAD-gated runtime, and on-device templates.

** not just accurate –
robust by design*



BEYOND SOTA:
What SoloSpeak Adds



1. Efficient KWS Models
(e.g., BC-ResNet, DS-CNN)
SOTA: extremely efficient keyword detection
SoloSpeak: *efficient keyword detection + enrolled-speaker verification*



2. Speaker Verification Systems
(x-vector, ECAPA-TDNN, etc.)
SOTA: strong speaker embeddings
SoloSpeak: *speaker identity is not enough – phrase and speaker must agree*



3. Personalized KWS Research
(task-adaptive / multi-task models)
SOTA: combines KWS + speaker verification
SoloSpeak: *adds disentanglement, hard-impostor mining, hysteresis, VAD gating, and Stage 7 export for real on-device deployment*

Resulting Strengths



Impostor rejected



Similar phrase rejected



Noise / TV / background rejected



Stable wake decision



Private on-device templates



Efficient & battery friendly



Validated Stage 7 Model

Clean TA
~93.97%

Wrong-speaker
rejection ~95.17%

Wrong-word
rejection ~97.83%

Background rejection
100%

External FA
~0.30%



*built for
real world*

Prototype Built. Stage 7 Deployable.

SoloSpeak is already trained, evaluated, calibrated, and exported as a working personalized wake-word prototype.

not just
a concept.

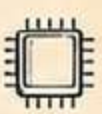
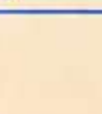


Our prototype journey

	Kaggle GPU Training trained with GPU resources	✓
	7-Stage Reliability Curriculum built layer-by-layer for robustness	✓
	Stage 7 Corrected Checkpoint final calibrated and improved model	✓
	Enrollment + Inference Runtime on-device enrollment and matching	✓
	Deployable Export Package ONNX export, quantization-ready	✓
	Codebase Available for Review open-source for transparency	✓

built → tested → exported

Achieved Prototype Metrics

	Model size	~1.10M parameters
	Clean True Accept (TA)	~93.97%
	Wrong-speaker rejection	~95.17%
	Wrong-word rejection	~97.83%
	Background rejection	100%
	External false accept (FA)	~0.30%
	External FA test scale	120 / 40,000
	Final threshold (τ)	$\tau \approx 0.27$

✓ Built · Trained · Evaluated · Exported · Prototype-ready



GitHub:

<https://github.com/PranavAngrish/Samsung-Designing-a-Robust-system-for-Speech-Disentanglement>

code open.
built with
privacy first.



Thank You

phrase +
speaker verified

voice identity

private
by design

one model,
many devices

 Team SoloSung · Pranav Angrish



<https://github.com/PranavAngrish/Samsung-Designing-a-Robust-system-for-Speech-Disentanglement>